



NAUVAL ZULFIKAR

NLP & AI Researcher · LLM Security and Safety Candidate Bandung, Indonesia (relocatable to Copenhagen, 2026-09-01) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264
[LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Visa status:** UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

NLP and AI researcher with a focus on LLM security and safety, privacy in NLP/AI, and robustness analysis across multilingual settings. **MSc Business Analytics, Aston University, 1:1 First Class**. Dissertation on blockchain-integrated LLM analysis for supply-chain customer reviews, supervised by Dr. Ammar Al-Bazi. Seeking to contribute to research on LLM security and safety under Prof. Johannes Bjerva at Aalborg University Copenhagen.

EDUCATION

Aston University, Birmingham, UK (*MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024*)

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan (*BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019*)

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- Distinctions: Business Data Analysis, Consumer Behaviour, Marketing Research

Indonesia AI (*NLP Engineer Training · EQF Level 5 · May – Nov 2023*)

- Contributed to the *2024 Indonesia Election Aspect-Based Sentiment Analyser* project
- Distinctions: Python Programming, Text Summarisation, Voice Recognition

Qarir Generator (Dubai, UAE) (*Data Science Global Accelerator · EQF Level 5 · Apr – Jun 2023*)

- Led sentiment-analysis development for the *Pfizer Public Sentiment Analysis* project
- Distinctions: Python, SQL, Pandas/NumPy

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules (2026)

- Developed an end-to-end RAG system using gpt-4o-mini for adaptive shipper decision rules
- github.com/nauvalZulfikar/LLM-Shipper-Profiles

Crowd-Shipping ABM Behavioural Sweep (2026)

- Conducted a 450-run parameter sweep with literature-anchored results
- github.com/nauvalZulfikar/Crowd-Shipping-ABM-

VRPOD MILP Optimisation (2026)

- MILP formulation for VRPOD problem, achieving 38.5% cost reduction
- github.com/nauvalZulfikar/VRP-MILP

Mini Digital Twin for Integrated Passenger-Freight Transport (2026)

- Built a three-layer Digital Twin prototype with live dashboards
- github.com/nauvalZulfikar/Mini-Digital-Twin

MSc Dissertation Pipeline (DeBERTa-v3 fine-tune) (2024)

- Fine-tuned DeBERTa-v3 for multi-tier supplier review data analysis

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung (*Data Scientist & Project Manager · Sep 2022 – Present*)

- Automated ETL pipelines and built dashboards, achieving 45% faster project delivery

Syncwell, Birmingham, UK (*Marketing Data Scientist · Jul – Sep 2024*)

- Implemented predictive ML and personalisation strategies, increasing CTR by 40%

PCOS Challenge, Birmingham, UK (*Strategic Data Scientist Specialist · Jun – Aug 2024*)

- Developed customer attribution models, reducing manual workload by 80%

Bank Muamalat Indonesia, Jakarta (*Lead Data Scientist · Nov 2019 – Aug 2022*)

- Developed NLP models for credit portfolio analysis, achieving a 5.3% investment return

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Comparative evaluation of multiple ML regressors for retail sales forecasting accuracy.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API + RAG, sentence-transformers, LangChain

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit

Mathematical/Formal: MILP (Pyomo, scipy.optimize.milp, HiGHS), Agent-Based Modelling, Discrete-Event Simulation, Digital Twin architecture, Bayesian methods

Governance: Model validation, model reliability, audit-grade documentation, A/B testing, AWS, Docker, Git, ETL/orchestration

LANGUAGES

Indonesian (*Native*) · **English** (*Fluent, IELTS 7.0 overall*) · **Japanese** (*Intermediate*)

REFEREES

Dr. Ammar Al-Bazi

Senior Lecturer, Aston Business School, Aston University

Email: a.al-bazi@aston.ac.uk

Relationship: MSc dissertation supervisor

Dr. Viktor Pekar

Lecturer, Aston Business & Social Sciences

Email: v.pekar@aston.ac.uk

Relationship: MSc Personal Tutor; taught Machine Learning for Business Analytics + Research Methods